

AUTONOMOUS SHUTTLE SERVICE : COLLEGE VENTURE PROJECT [SRMIST]

Vivek Ranjan¹, Sidhant Dhall²

SRM University, Department of Electronics & Instrumentation, Chennai-603203, India

E-mail- ¹vivek.ranjan.srm@gmail.com, ²sidhant.dhall@gmail.com

Ph.No. - ¹+918015215390, ²+919840197142

Abstract:

A line follower is an autonomous robot that moves by sensing a particular line. Typically, IR or ULTRASONIC sensors are fixed on the body of the robot. These sensors trace the line and send signals to a micro-controller which is programmed to control the motion of the robot and move it along the line.

A line follower circuit can be represented by a simple block diagram as follows :-

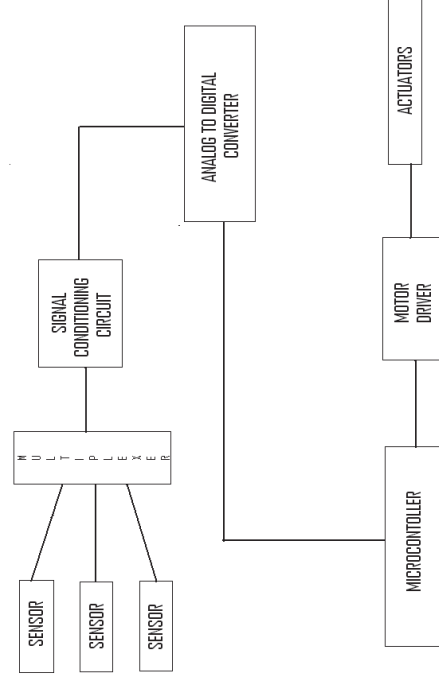
Now based on this simple principle of a line follower, we propose to develop an autonomous

shuttle for our own campus. The shuttle will act as a supplement to our present shuttle service.

Existing System:

After a crude survey on the present existing shuttle service, we have found out that everyday the college spends Rs.6500/- on an average for the running and maintenance of the vehicles. The college runs 2 big buses and 1 small shuttle inside our campus. A brief statistics about the shuttles can be given as follows :-

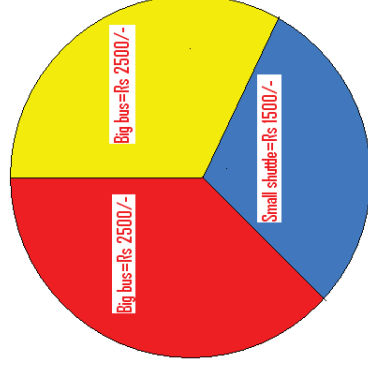
1. Buses :- Rs. 2500/- per bus (including



fuel and manual charges)

2. 1 small shuttle :- Rs. 1500 (including fuel and manual charges)

So, the total charges come around :- Rs(2 * 2500 + 1500) = **Rs. 6500/- daily.**



This cost is too high. And we propose that by making our autonomous shuttle we can reduce our cost substantially.



Proposal:

The shuttle that we propose will be completely autonomous and will require no manual labour. The shuttle will work on electric supply and will not take any kind of fuel. A thin metal track will be laid on the roads of our campus. The shuttle will trace the metal track as in a line follower.

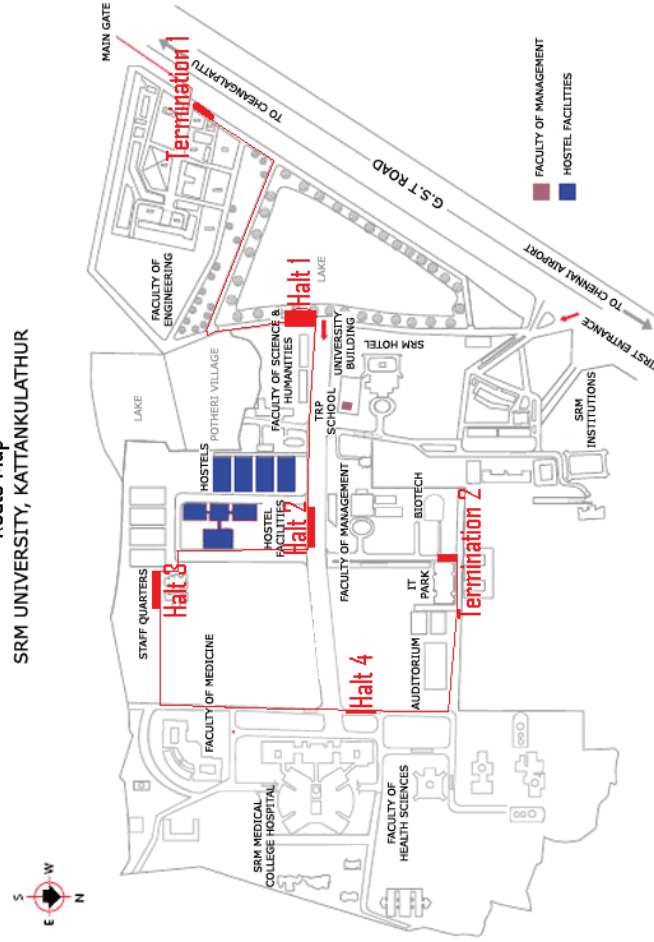
like solar energy and alternator principles.

As a pilot scale, we are planning to make it a ten carriage shuttle. This can be extended further to more people.

Our proposed map is given as below :-

This shuttle will start from the front gate of the main campus and will cover the

Route Map
SRM UNIVERSITY, KATTANKULATHUR



We plan to make it an autonomous electric vehicle which will also use alternate sources of energy for power

entire campus including main campus, the hostels, staff quarters, SRM hospital, medical campus, main auditorium and the IT park.

Metal sensors will be fixed under the body of the shuttle. These sensors will sense the metal plate fixed on the ground and the shuttle will move accordingly following the line.

➤ Track System :-

A thin metal plate will be laid linearly along the whole path of the shuttle. The track will be completely labeled on the ground so that it does not cause any disturbance in the present traffic in the campus.

The track will be of a material of high tensile strength with low coefficient of expansion. We will take care that the track will not be corroded anyway.

➤ Batteries :-

Lead acid batteries will be the major power source in the shuttle. A battery bank will be installed on the shuttle which will give the required voltage required by the motors.

➤ Obstacle avoiding system :-

The shuttle will also be fitted with Ultrasonic sensors that can sense any living or obstacle in its path. It will be fitted with an efficient obstacle avoiding system that will control the vehicle in any such case.

➤ Alternate Power :-

The shuttle will use alternate energy sources like solar cells and other dynamo principles to charge the batteries. This system will suffice the

Components we plan to use :-

➤ Micro-Controllers :-

The shuttle will be completely controlled by micro-controllers. Different micro-controllers will be used for the metal tracing and the obstacle avoiding systems. The micro-controllers will give the shuttle the unique feature of being completely autonomous. These micro-controllers will be pre-programmed to stop the shuttle at the requisite halts and wait for the passengers to embark and disembark and then start the shuttle after a fixed interval of time. The micro-controllers will also decide the way taken by the shuttle.

➤ DC series Motors :-

There will be 4 motors and each of them will be connected to the 4 wheels. The motors can provide high torque initially while starting the vehicle thus providing the required acceleration. Then the motor will run at a constant speed once it reaches saturation making the shuttle run at a constant speed with very good stability.

➤ SERVO Motor :-

A servo motor will be controlled using a micro-controller. Using this motor, we will make a steering mechanism. The servo motor will be connected to a axle of the front wheel.

➤ Metal detectors :-

normal charging of the batteries by supplying a AC supply.

We are working to use other types of alternate sources of energy to make the vehicle more eco-friendly.

➤ Chassis of the Shuttle :-

The shuttle will be made of a ultra light material like hard plastics. The shuttle will be covered by fibre glass. These materials will make the weight of the body very less and we can maximize its efficiency this way. Solar panels will be fixed on the roof of the

shuttle. These panels will collect solar energy throughout the day and this energy will be used to charge the batteries.

Below are a few schematics of the various systems in the shuttle :-

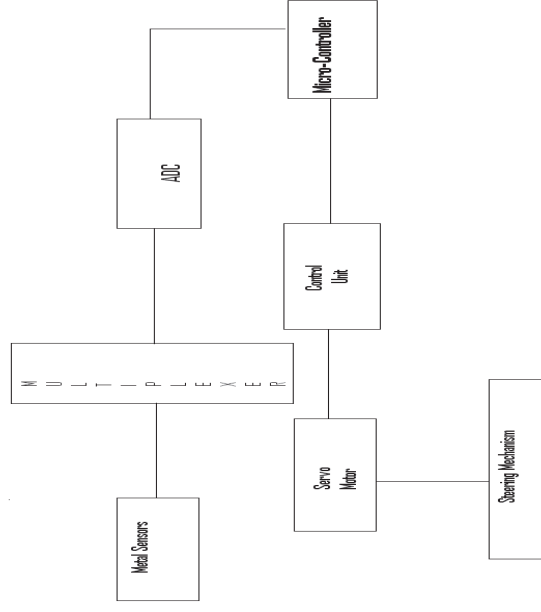


Fig. 1 - The Steering mechanism of the shuttle

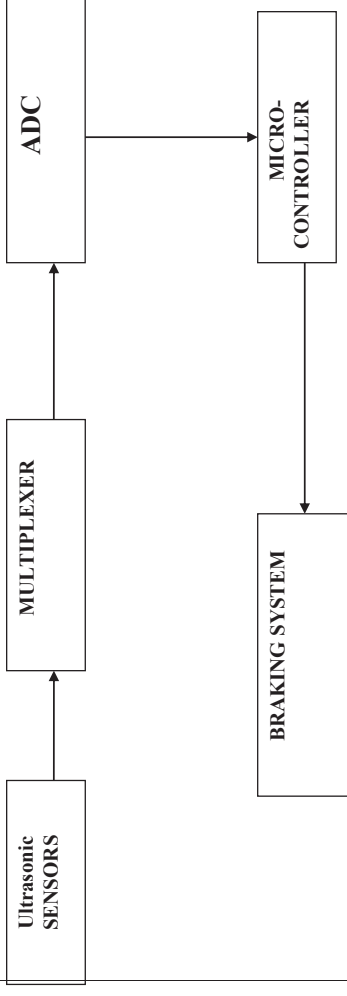


Fig :- Obstacle avoiding system of the shuttle.

- The shuttle will be eco-friendly. It will cause no pollution in the

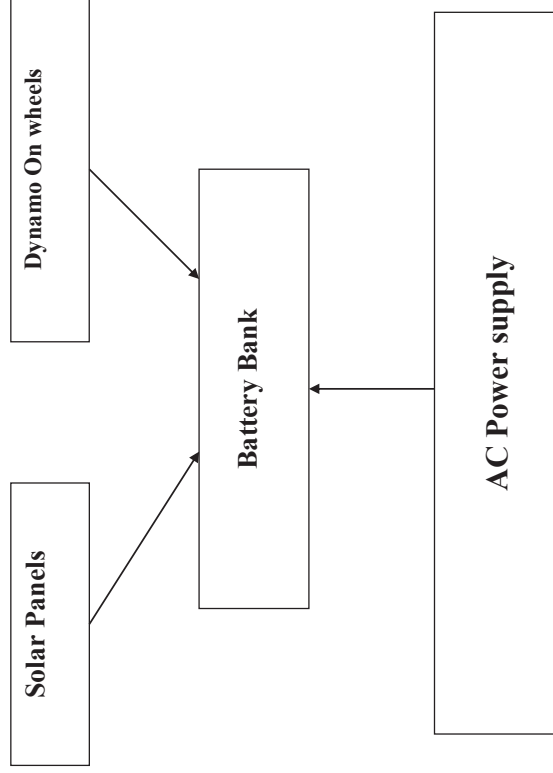


Fig :- Charging of the battery in the shuttle

Advantages:

The advantages of this shuttle can be enormous. A few of the advantages are given below :-

- The shuttle will operate perfectly on time and hence make the waiting time for students very less.
- The shuttle will be unique of its kind.
- The shuttle can bring down the cost of the present shuttle service drastically.

- The shuttle can, in due course of time, replace the entire shuttle service with a network of such shuttles running inside our campus.
- The shuttle will make our campus one of the most hi-tech campus in the country.
- It is a small but significant step towards artificial intelligence.

Conclusion :

After detailing the terminology of the entire system proposed to be made is practically proved to be feasible and can all round be capable to cover the track with the help of GPS and the path following method avoiding the static obstacles that would be coming in its track. It can adapt to situation changes in case of faulty responses and errors and can be well monitored on the signaling PC for further assistance. Thus the project is worked on to challenge all the tasks and complete it with least errors and maximum efficiency.

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